DECENT COACHING CLASSES

X CBSE | Mind Your Math - II — ANSWER KEY

Section A Answers

1. (b) $a \times b$ 2. (a) $x^2 - 4x + 3$ 3. (c) infinitely many solutions 4. (b) perpendicular 5. (a) 2/5

Section B Answers

1. $96 = 2^5 \times 3$, $120 = 2^3 \times 3 \times 5$. HCF $= 2^3 \times 3 = 24$. LCM $= 2^5 \times 3 \times 5 = 480$

2. Sum of zeroes $= 5 + (-3) = 2$. Product of zeroes $= 5 \times (-3) = -15$. Required polynomial: $x^2 - 2x - 15$

3. $a_1/a_2 = 2/4 = 1/2$, $b_1/b_2 = -3/-6 = 1/2$, $c_1/c_2 = 7/9$. Since $a_1/a_2 = b_1/b_2 \neq c_1/c_2$, the lines are parallel and the pair of equations is inconsistent

4. Since the tangent is perpendicular to the radius at the point of contact, ΔOPQ is right-angled at P. $PQ^2 = OQ^2 - OP^2 = 10^2 - 6^2 = 100 - 36 = 64$. ∴ $PQ = 8$ cm

5. Area $= (\theta/360) \times \pi r^2 = (90/360) \times 22/7 \times 14^2 = 1/4 \times 22/7 \times 196 = 154$ cm^2

Section C Answers

1. Assume, to the contrary, that $\sqrt{5}$ is rational. Then $\sqrt{5} = a/b$ for coprime integers a, b ($b \neq 0$). So $5b^2 = a^2$, which means 5 divides a^2, and hence 5 divides a. Let $a = 5c$. Substituting, $5b^2 = 25c^2$, so $b^2 = 5c^2$, which means 5 divides b^2, and hence 5 divides b. So a and b have 5 as a common factor, contradicting that they are coprime. Hence $\sqrt{5}$ is irrational.

2. Comparing $2x^2 - 5x + 3$ with $ax^2 + bx + c$: $a = 2$, $b = -5$, $c = 3$. $\alpha + \beta = -b/a = 5/2$, $\alpha\beta = c/a = 3/2$. $\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta = (5/2)^2 - 2(3/2) = 25/4 - 3 = 13/4$

3. Multiplying $3x + 2y = 11$ by 3 and $2x + 3y = 4$ by 2: $9x + 6y = 33$ and $4x + 6y = 8$. Subtracting: $5x = 25 \Rightarrow x = 5$. Substituting in $3x + 2y = 11$: $15 + 2y = 11 \Rightarrow 2y = -4 \Rightarrow y = -2$

Section D Answer

(a) The cow can graze in a quarter circle of radius 7 m (angle 90° at the corner). Area $= (90/360) \times \pi r^2 = 1/4 \times 22/7 \times 7^2 = 1/4 \times 22/7 \times 49 = 38.5$ m^2

(b) Total outcomes = 52. (i) Number of face cards = 12. P(face card) = 12/52 = 3/13. (ii) Number of red kings = 2. P(red king) = 2/52 = 1/26